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Surgical access device having plural zero closure valves

Abstract

A surgical access device includes a seal assembly, an upper housing portion, a lower housing portion, and a tubular member. The seal assembly has an instrument seal and is coupled to the upper housing portion. A first valve is partially disposed in an upper chamber of the upper housing portion and a second valve is partially disposed in a lower chamber of the lower housing portion. A valve assembly extends from the upper housing portion and is in fluid communication with the upper chamber. A source of insufflation fluid is attachable to the valve assembly.

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Background/Summary

FIELD

(1) The present disclosure generally relates to surgical instruments for accessing a surgical site. In particular, the present disclosure relates to a surgical access device having plural zero closure valves.

BACKGROUND

(2) In minimally invasive surgical procedures, including endoscopic and laparoscopic surgeries, a surgical access device permits the introduction of a variety of surgical instruments into a body cavity or opening. A surgical access device (e.g., a cannula or an access port) is introduced through an opening in tissue (e.g., a naturally occurring orifice or an incision) to provide access to an underlying surgical site in the body. The opening is typically made using an obturator having a blunt or sharp tip that may be inserted through a passageway of the surgical access device. For example, a cannula has a tube of rigid material with a thin wall construction, through which an obturator may be passed. The obturator is utilized to penetrate a body wall, such as an abdominal wall, or to introduce the surgical access device through the body wall, and is then removed to permit introduction of surgical instruments through the surgical access device to perform the minimally invasive surgical procedure.

(3) Minimally invasive surgical procedures, including both endoscopic and laparoscopic procedures, permit surgery to be performed on organs, tissues, and vessels far removed from an opening within the tissue. In laparoscopic procedures, the abdominal cavity is insufflated with an insufflation gas, e.g., CO.sub.2, to create a pneumoperitoneum thereby providing access to the underlying organs. A laparoscopic instrument is introduced through a cannula into the abdominal cavity to perform one or more surgical tasks. The cannula may incorporate a seal to establish a substantially fluid tight seal about the laparoscopic instrument to preserve the integrity of the pneumoperitoneum. The cannula, which is subjected to the pressurized environment, e.g., the pneumoperitoneum, may include an anchor to prevent the cannula from backing out of the opening in the abdominal wall, for example, during withdrawal of the laparoscopic instrument from the cannula.

(4) Recently, laparoscopic peritoneum escape ("LPE") has become a greater operating room

concern due to a potentially contaminated patient could infect operating room personnel during or after a surgical procedure as fluids (i.e., liquids or gases) escape the abdomen due to leaks around surgical instruments during insertion and/or removal of the surgical instruments through a surgical access device.

SUMMARY

(5) A surgical access device includes a seal assembly having an instrument seal therein and an upper housing portion coupled to the seal assembly. The upper housing portion includes a first valve that is partially disposed in an upper chamber of the upper housing portion. A lower housing portion is attached to the upper housing portion and includes a second valve that is partially disposed in a lower chamber of the lower housing portion. A valve assembly extends from the upper housing portion and is in fluid communication with the upper chamber. The valve assembly is attachable to a source of insufflation fluid. A tubular member extends from the lower housing portion.

(6) In aspects, an open position of the valve assembly may be configured to introduce insufflation fluid into the upper chamber and define a first pressure therein.

(7) In an aspect, the first pressure may be greater than an ambient pressure of the seal assembly thereby defining a first differential pressure that is capable of urging the first valve towards a closed configuration.

(8) In one aspect, the first pressure may be greater than a second pressure of the lower chamber thereby defining a second differential pressure that is capable of urging the second valve towards an open configuration.

(9) In aspects, a surgical instrument may be inserted through the seal assembly and the first valve. The first differential pressure may be capable of maintaining the first valve in contact with the surgical instrument.

(10) In further aspects, an open position of the valve assembly may be configured to introduce insufflation fluid into the upper chamber and define a first pressure therein. The first pressure may be greater than an ambient pressure in the seal assembly thereby defining a first differential pressure. The first pressure may be greater than a second pressure in the lower chamber thereby defining a second differential pressure. The first differential pressure may be capable of urging the first valve towards a closed position and the second differential pressure may be capable of urging the second valve towards an open position.

(11) In aspects, one of the first or second valves may be a duckbill valve.

(12) In an aspect, the seal assembly may be removably coupled to the upper housing portion.

(13) In another aspect of the present disclosure, a surgical access system has a housing including first and second housing portions with respective first and second chambers. A first valve is partially disposed in the first chamber and a second valve is partially disposed in the second chamber. A valve assembly is connected to the first housing portion and couples a source of fluid with the first chamber. The first chamber has a first pressure with the valve assembly in an open position. A seal assembly is coupled to the first housing portion and has an instrument seal and an ambient pressure therein. The ambient pressure is less than the first pressure thereby defining a first differential pressure that urges the first valve towards a closed position. A tubular member extends from the second housing portion.

(14) In aspects, the first pressure may be greater than a second pressure of the second chamber thereby defining a second differential pressure that urges the second valve towards an open position.

(15) In other aspects, a surgical instrument may be inserted through the seal assembly and the first valve, wherein the first differential pressure maintains contact between the first valve and the surgical instrument.

(16) In an aspect, the first pressure may be greater than a second pressure in the second chamber thereby defining a second differential pressure that urges the second valve towards an open

position.
(17) In an aspect, one of the first or second valves may be a duckbill valve.
(18) In one aspect, the seal assembly may be removably coupled to the first housing portion.
(19) In further aspects, the first pressure may be greater than a second pressure in the second chamber thereby defining a second differential pressure that urges the second valve towards an open position thereby facilitating insertion of the surgical instrument through the second valve.
(20) In another aspect of the disclosure, a method of coupling a surgical instrument and a surgical access device includes inserting a surgical instrument into a surgical access device where the surgical access device includes a housing with first and second housing portions and respective first and second chambers. A first valve is partially disposed in the first chamber, a second valve is partially disposed in the second chamber, and a valve assembly is attached to the first housing portion. The valve assembly is fluidly coupled to a source of fluid. A seal assembly is coupled to the first housing portion, and a tubular member extends from the second housing portion. The method also includes opening the valve assembly thereby introducing fluid to the first chamber and defining a first pressure therein. The first pressure is greater than an ambient pressure of the seal assembly thereby defining a first differential pressure that transitions the first valve to a closed position. The first pressure is greater than a second pressure in the second chamber thereby defining a second differential pressure that transitions the second valve to an open position. Additionally, the method includes advancing the surgical instrument through the first and second valves where the first differential pressure maintains contact between the surgical instrument and the first valve. The method also includes withdrawing a distal portion of the surgical instrument into the first chamber where the first differential pressure maintains contact between the surgical instrument and the first valve, while the second differential pressure maintains the second valve in the open position. Further, the method includes withdrawing the distal portion of the surgical instrument into the seal assembly, where the first differential pressure transitions the first valve to the closed position.
(21) In an aspect, inserting the surgical instrument into the surgical access device may include removably coupling the seal assembly to the first housing portion.
(22) In aspects, advancing the surgical instrument through the first and second valves may include inserting the surgical instrument through at least one duckbill valve.
(23) In a further aspect, advancing the surgical instrument through the first and second valves may include inserting the surgical instrument through first and second duckbill valves.
(24) Other features of the disclosure will be appreciated from the following description.

Description

DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate aspects and features of the disclosure and, together with the detailed description below, serve to further explain the disclosure, in which:
(2) FIG. **1** is a perspective view of a surgical access device according to an aspect of the present disclosure;
(3) FIG. **2** is an exploded perspective view, with parts separated, of the surgical access device of FIG. **1**;
(4) FIG. **3** is a side cross-sectional view of the surgical access device of FIG. **1** taken along section line **3-3**;
(5) FIG. **4** is a perspective view of the surgical access of FIG. **1** with a surgical instrument inserted therethrough and positioned in body tissue;
(6) FIG. **5** is a side cross-sectional view of the surgical access device of FIG. **4** taken along section

line **5-5** with the surgical instrument engaging first and second zero closure valves;
(7) FIG. **6** is a side cross-sectional view of the surgical access device of FIG. **5** with the surgical instrument partially withdrawn and engaging the first zero closure valve; and
(8) FIG. **7** is a side cross-sectional view of the surgical access device of FIG. **5** with the surgical instrument partially withdrawn and disengaged from the first and second zero closure valves.
DETAILED DESCRIPTION
(9) Aspects of the disclosure are described hereinbelow with reference to the accompanying drawings; however, it is to be understood that the disclosed aspects are merely exemplary of the disclosure and may be embodied in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosure in virtually any appropriately detailed structure.
(10) Descriptions of technical features of an aspect of the disclosure should typically be considered as available and applicable to other similar features of another aspect of the disclosure. Accordingly, technical features described herein according to one aspect of the disclosure may be applicable to other aspects of the disclosure, and thus duplicative descriptions may be omitted herein. Like reference numerals may refer to like elements throughout the specification and drawings. For a detailed description of the structure and function of exemplary surgical access assemblies, reference may be made to U.S. Pat. Nos. 7,300,448; 7,691,089; and 8,926,508, the entire content of each of which is hereby incorporated by reference herein.
(11) With initial reference to FIGS. **1-3**, a surgical access device **100** is illustrated. The surgical access device **100** has a seal assembly **10**, a housing **30**, and a tubular member or cannula **50**. The housing **30** includes an upper or first housing portion **32** and a lower or second housing portion **34**. The seal assembly **10** is releasably coupled to a proximal portion of the upper housing portion **32** and the cannula **50** extends from the lower housing portion **34**. The cannula **50** includes a tubular member **52** and may include ribs or other protrusions **54** along a portion of its length that help stabilize the cannula **50** when it is inserted into tissue (FIG. **4**). The seal assembly **10** has a proximal opening **22** and a distal opening **24** (FIG. **3**) defining a passage **26** through the seal assembly **10**. The seal assembly **10** further includes a base **12** and a body **14** coupled to the base **12**. An instrument seal **40** is positioned in the seal assembly **10** and includes a central orifice **42** that sealingly engages a surgical instrument I (FIG. **4**) inserted through the passage **26** of the seal assembly **10**. When the surgical instrument I is inserted through the central orifice **42**, it is engaged with the instrument seal **40** and the instrument seal **40** provides a fluid-tight barrier. The upper housing portion **32** has an upper or first chamber **36** defined therein and a valve assembly **70** extending radially from an outer surface of the upper housing portion **32**. The lower housing portion **34** extends distally from the upper housing portion **32** and has a lower or second chamber **38** defined therein. The seal assembly **10**, the upper housing portion **32**, the lower housing portion **34**, and the cannula **50** are formed from a suitable biocompatible polymeric material (e.g., polycarbonate).
(12) A first valve **110** is disposed in the upper housing portion **32**. In particular, a flange **116** of the first valve **110** engages a rim **37** of the upper housing portion **32** thereby supporting the first valve **110** in the upper housing portion **32**. The first valve **110** is a conical elastomeric membrane, such as a duckbill or zero-closure valve fabricated from a resilient material, such as, for example, rubber, etc. The first valve **110** is flexible for resilient reception of the surgical instrument I and maintaining seal integrity between the first chamber **36** and the ambient environment. The first valve **110** includes first and second flaps **112***a*, **112***b* that are inwardly biased towards a contact region **114** so as to form a seal. First and second flaps **112***a*, **112***b* define the contact region **114** that is configured for receiving the surgical instrument I therethrough. The contact region **114** permits passage of the surgical instrument I through the first valve **110** whereby the first and second flaps

112*a*, **112***b* form a substantial seal with a shaft of the surgical instrument I when it is inserted therethrough. In the absence of the surgical instrument I being inserted through the first valve **110**, the contact region **114** forms a fluid tight seal that isolates the first chamber **36** from the ambient environment.

(13) A second valve **130** is disposed in the lower housing portion **34**. In particular, a flange **136** of the second valve **130** engages a groove **39** of the lower housing portion **34** thereby supporting the second valve **130** in the lower housing portion **34**. The second valve **130** is a conical elastomeric membrane, such as a duckbill or zero-closure valve fabricated from a resilient material, such as, for example, rubber, etc. The second valve **130** is flexible for resilient reception of the surgical instrument I and maintaining seal integrity between the first chamber **36** and the second chamber **38**. The second valve **130** includes first and second flaps **132***a*, **132***b* that are inwardly biased towards a contact region **134** so as to form a seal. First and second flaps **132***a*, **132***b* define the contact region **134** that is configured for receiving the surgical instrument I therethrough. The contact region **134** permits passage of the surgical instrument I through the second valve **130** whereby the first and second flaps **132***a*, **132***b* form a substantial seal with a shaft of the surgical instrument I when it is inserted therethrough. In the absence of the surgical instrument I being inserted through the second valve **130**, the contact region **134** forms a fluid tight seal that isolates the second chamber **38** from a lumen **56** (FIG. **4**) of the cannula **50**.

(14) With reference to FIGS. **3-5**, the ambient air surrounding the surgical access device **100** has an ambient pressure Pa that is exerted on an exterior of the surgical access device **100**. At sea level, the ambient pressure is approximately 14.7 psi or 101.325 kPa. A first differential pressure (ΔP) is defined by the difference between the ambient pressure P.sub.a and a first pressure P.sub.1 of the first chamber **36**. If the first pressure P.sub.1 is greater than the ambient pressure P.sub.a, then the first ΔP is positive. A negative first ΔP exists when the ambient pressure P.sub.a is greater than the first pressure P.sub.1. Similarly, a second ΔP is defined by the difference between the first pressure P.sub.1 and a second pressure P.sub.2 of the second chamber **38**, which is substantially equal to a pressure present in the lumen **56** of the cannula **50**. If the second pressure P.sub.2 is greater than the first pressure P.sub.1, then the second ΔP is positive. A negative second ΔP exists when the first pressure P.sub.1 is greater than the second pressure P.sub.2. In the initial state, the ambient pressure P.sub.a, the first pressure P.sub.1, and the second pressure P.sub.2 are all substantially equal and the first and second valves **110**, **130** remain in a closed configuration.

(15) The base **12** of the seal assembly **10** is releasably attached to the upper housing portion **32** of the housing **30**. A tab **16** is integrally formed with the seal assembly **10** and is configured for resilient movement relative to the upper housing portion **32** such that the tab **16** is movable in a distal direction relative to the seal assembly **10**. Moving the tab **16** distally permits a user to rotate the seal assembly **10** relative to the upper housing portion **32** for removal of the seal assembly **10** from the surgical access device **100**. Attachment of the seal assembly **10** to the upper housing portion **32** of the surgical access device **100** involves rotating the seal assembly **10** relative to the upper housing portion **32** in an opposite direction. An example of a seal assembly coupled to a housing of a surgical access device is described in commonly owned U.S. Pat. No. 10,022,149, the entire content of which is incorporated herein by reference.

(16) The valve assembly **70** includes a body **72**, a valve handle **74**, and a port **76**. The valve assembly **76** may be a stopcock style valve. The valve handle **74** is rotatable between an open position (FIG. **4**) that allows fluid flow into or out of the valve assembly **70** and a closed position (FIG. **1**) that inhibits fluid flow into or out of the valve assembly **70**. The port **76** may have a fitting such as a luer fitting and is configured to allow the valve assembly **70** to be coupled to a source of insufflation fluid FS via tube T that has an insufflation pressure P.sub.i that is greater than the ambient pressure P.sub.a. When the pressurized insufflation fluid is introduced into the first chamber **36** via the valve assembly **70**, the first pressure P.sub.1 of the first chamber is also greater than the ambient pressure P.sub.a (i.e., positive first ΔP) thereby maintaining the first valve **110** in

the closed configuration. This further limits any fluids that might escape from the surgical access assembly **100** through the first valve **110** to the environment surrounding the surgical access device **100** (e.g., operating room). Additionally, the first pressure P.sub.1 of the first chamber **36** is now greater than the second pressure P.sub.2 of the second chamber **38** and the second ΔP is negative thereby urging the second valve **130** towards an open configuration and allowing the insufflation fluid to flow into the second chamber **36** through the lumen **56** of the cannula **50** and into the body cavity BC (e.g., a surgical working space) as indicated by the arrows A.
(17) Referring now to FIGS. **5-7**, insertion and removal of the surgical instrument I with respect to the surgical access device **100** are illustrated. Initially, as seen in FIG. **5**, the surgical instrument I is inserted through the proximal opening **22** (FIG. **4**) of the seal assembly **10**, through the central orifice **42** of the instrument seal **40**, through the contact regions **114**, **134** of the first and second valves **110**, **130**, and into the lumen **56** of the cannula **50**. The resilient nature of the flaps **112***a*, **112***b* of the first valve **110** maintain contact between the flaps **112***a*, **112***b* and an outer surface of the surgical instrument I thereby defining a fluid-tight boundary. The fluid-tight boundary between the first valve **110** and the surgical instrument I is augmented by the insufflation fluid entering the first chamber **36** from the valve assembly **70**. As the insufflation fluid enters the first chamber **36**, the first chamber **36** has a higher pressure than the ambient pressure P.sub.a in the seal assembly **10** and the positive first ΔP urges the flaps **112***a*, **112***b* of the first valve **110** towards the closed configuration thereby maintaining contact between the flaps **112***a*, **112***b* and the surgical instrument I. This arrangement inhibits fluids (e.g., liquids or gases) from entering the seal assembly **10** from the first chamber **36**. Additionally, the first pressure P.sub.1 in the first chamber **36** is greater than the second pressure P.sub.2 in the second chamber **38** and the lumen **56** of the cannula **50** thereby defining the negative second ΔP. The negative second ΔP urges the flaps **132***a*, **132***b* of the second valve **130** towards the open configuration. As such, the insufflation fluid flows along the outer surface of the surgical instrument I and through the contact region **134** of the second valve **110** and into the lumen **56** of the cannula **50** as shown by arrows A. This flow path further limits any fluids from exiting the surgical access device **100** through the proximal opening **22** of the seal assembly **10**.
(18) In FIG. **6**, the surgical instrument I is being withdrawn in the direction of arrow B from the surgical access device **100** while insufflation fluid is still being supplied to the surgical access device **100** through the valve assembly **70** as shown by arrows A. Similar to the arrangement in FIG. **5**, the flaps **112***a*, **112***b* of the first valve **110** are in contact with the outer surface of the surgical instrument I and are urged towards the closed configuration by the positive first ΔP thereby maintaining contact between the flaps **112***a*, **112***b* and the surgical instrument I to inhibit any fluids exiting the surgical access device **100** through the seal assembly **10**. Additionally, the negative second ΔP keeps the flaps **132***a*, **132***b* of the second valve **130** from closing thereby allowing fluid flow from the first chamber **36** through the second valve **130** and into the second chamber **38** as well as the lumen **56** of the cannula **50** as shown by arrows A. The instrument seal **40** also maintains contact with the surgical instrument I (FIGS. **5** and **6**) to inhibit any fluid flow past the instrument seal I towards the proximal opening **22**.
(19) As seen in FIG. **7**, the surgical instrument I has been withdrawn to a point where a distal tip of the surgical instrument I is positioned proximally of the instrument seal **40** in the seal assembly **10**. As such, the instrument seal **40** no longer inhibits fluid flow in either direction through the central orifice **42** of the instrument seal **40**. As the valve handle **74** of the valve assembly **70** is still in the open position, insufflation fluid is still being supplied to the surgical access device **100** as shown by arrows A. In particular, the insufflation fluid enters the first chamber with a higher pressure P.sub.i than the ambient pressure P.sub.a in the seal assembly **10** maintaining the positive first ΔP. The positive first ΔP continues to urge the flaps **112***a*, **112***b* of the first valve **110** towards a closed configuration augmenting the flaps **112***a*, **112***b* bias towards the closed configuration. In combination, the bias of the flaps **112***a*, **112***b* and the positive first ΔP inhibit proximal fluid flow

across the first valve **110**. Concurrently, the first pressure P.sub.1 in the first chamber **36** is greater than the second pressure P.sub.2 in the second chamber **38** and maintains the negative second ΔP. The negative second ΔP urges the flaps **132***a*, **132***b* of the second valve **130** apart allowing insufflation fluid to flow from the first chamber **36**, through the second valve **130**, into the second chamber **38**, and into the lumen **56** of the cannula **50**. The positive pressure P.sub.1 of the first chamber **36** relative to the ambient pressure P.sub.a inhibits LPE of fluids in the surgical access device **100**. Further, since the pressure P.sub.1 in the first chamber **36** is greater than the pressure P.sub.2 in the second chamber **38**, which is equal to the pressure in the lumen **56** of the cannula **50**, any LPE of fluids is directed towards the body cavity BC (e.g., surgical access site) rather than potentially escaping into the environment surrounding the surgical access device **100** (e.g., operating room).

(20) Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting. It is envisioned that the elements and features may be combined with the elements and features of another without departing from the scope of the disclosure. As well, one skilled in the art will appreciate further features and advantages of the disclosure.

Claims

1. A surgical access device comprising: a seal assembly having an instrument seal; an upper housing portion coupled to the seal assembly; a first valve partially disposed in an upper chamber of the upper housing portion; a lower housing portion comprising a proximal end attached to the upper housing portion and comprising an opposite distal end, the lower housing portion comprising a conical neck narrowing an inner diameter of the lower housing portion from a first inner diameter at the proximal end to a second, smaller inner diameter at the distal end; a second valve partially disposed in a lower chamber of the lower housing portion and supported by the lower housing portion proximal of the conical neck, the first valve and the second valve being duckbill valves, the second valve spanning the first inner diameter and including a circumferential flange and first and second flaps that form a seal at a contact region; a valve assembly extending from the upper housing portion and in fluid communication with the upper chamber, the valve assembly attachable to a source of insufflation fluid; and a cannula extending from the lower housing portion, the cannula defining a lumen having a lumen diameter less than the first inner diameter of the lower housing portion, wherein the contact region forms a distal-most end of the second valve that is disposed proximal to the cannula when the circumferential flange is disposed between the upper housing portion and the proximal end of the lower housing portion.
2. The surgical access device of claim 1, wherein an open position of the valve assembly is configured to introduce insufflation fluid into the upper chamber and define a first pressure of the upper chamber.
3. The surgical access device of claim 2, wherein the first pressure is greater than an ambient pressure of the seal assembly thereby defining a first differential pressure that is capable of urging the first valve towards a closed configuration.
4. The surgical access device of claim 3, further including a surgical instrument inserted through the seal assembly and the first valve, wherein the first differential pressure is capable of maintaining the first valve in contact with the surgical instrument.
5. The surgical access device of claim 2, wherein the first pressure is greater than a second pressure of the lower chamber thereby defining a second differential pressure that is capable of urging the second valve towards an open configuration.
6. The surgical access device of claim 1, wherein an open position of the valve assembly is configured to introduce insufflation fluid into the upper chamber and define a first pressure of the upper chamber, the first pressure greater than an ambient pressure in the seal assembly thereby

defining a first differential pressure, the first pressure greater than a second pressure in the lower chamber thereby defining a second differential pressure, the first differential pressure capable of urging the first valve towards a closed position and the second differential pressure capable of urging the second valve towards an open position.

7. The surgical access device of claim 6, further including a surgical instrument inserted through the seal assembly and the first valve, wherein the first differential pressure is capable of maintaining the first valve in contact with the surgical instrument.

8. The surgical access device of claim 1, wherein the seal assembly is removably coupled to the upper housing portion.

9. The surgical access device of claim 1, wherein the first valve and the second valve are symmetric duckbill valves.

10. A surgical access system comprising: a housing including a first housing portion having a first chamber and a second housing portion having a second chamber, the second housing portion further comprising a proximal end and an opposite distal end and a conical neck narrowing an inner diameter of the second housing portion from a first inner diameter at the proximal end to a second, smaller inner diameter at the distal end; a first valve partially disposed in the first chamber; a second valve partially disposed in the second chamber, the first valve and the second valve being duckbill valves, the second valve supported by the second housing portion proximal of the conical neck, the second valve spanning the first inner diameter and including a circumferential flange and first and second flaps that form a seal at a contact region; a valve assembly connected to the first housing portion, the valve assembly coupling a source of fluid with the first chamber, the first chamber having a first pressure with the valve assembly in an open position; a seal assembly coupled to the first housing portion, the seal assembly having an instrument seal and an ambient pressure therein, the ambient pressure less than the first pressure thereby defining a first differential pressure that urges the first valve towards a closed position; and a cannula extending from the distal end of the second housing portion, the cannula defining a lumen having a lumen diameter less than the first inner diameter of the second housing portion, wherein the contact region forms a distal-most end of the second valve that is disposed proximal to the cannula when the circumferential flange is disposed between the first housing portion and the proximal end of the second housing portion.

11. The surgical access system of claim 10, wherein the first pressure is greater than a second pressure of the second chamber thereby defining a second differential pressure that urges the second valve towards an open position.

12. The surgical access system of claim 10, further including a surgical instrument inserted through the seal assembly and the first valve, wherein the first differential pressure maintains contact between the first valve and the surgical instrument.

13. The surgical access system of claim 12, wherein the first pressure is greater than a second pressure in the second chamber thereby defining a second differential pressure that urges the second valve towards an open position thereby facilitating insertion of the surgical instrument through the second valve.

14. The surgical access system of claim 10, wherein the seal assembly is removably coupled to the first housing portion.

15. A surgical access device comprising: a seal assembly having an instrument seal; a housing including an upper housing portion and a lower housing portion, the lower housing portion comprising a proximal end, a distal end, and a conical neck narrowing an inner diameter of the lower housing portion from a first inner diameter at the proximal end to a second, smaller inner diameter at the distal end; a first valve disposed in an upper chamber of the upper housing portion, the upper chamber having a first pressure; a second valve disposed in a lower chamber of the lower housing portion and supported by the lower housing portion proximal of the conical neck, the lower chamber having a second pressure, wherein when the second pressure is greater than the first

pressure, the second valve is urged towards a closed position, and wherein the second valve spans the first inner diameter and includes a circumferential flange and first and second flaps that form a seal at a contact region; a valve coupled to the upper housing portion and in fluid communication with the upper chamber, the valve attachable to a source of fluid; and a cannula extending from the lower housing portion, the cannula defining a lumen having a lumen diameter less than the first inner diameter of the lower housing portion, wherein the contact region forms a distal-most end of the second valve that is disposed proximal to the cannula when the circumferential flange is disposed between the upper housing portion and the proximal end of the lower housing portion.

16. The surgical access device of claim 15, wherein an open position of the valve is configured to introduce insufflation fluid into the upper chamber.

17. The surgical access device of claim 15, wherein the first pressure is greater than an ambient pressure thereby defining a first differential pressure that urges the first valve towards a closed position.

18. The surgical access device of claim 17, further including a surgical instrument inserted through the seal assembly and the first valve, wherein the first differential pressure is capable of maintaining the first valve in contact with the surgical instrument.

19. The surgical access device of claim 15, wherein the first valve and the second valve are proximal of the cannula.

20. The surgical access device of claim 15, wherein the first valve and the second valve are symmetric duckbill valves.